

Freeze ablation – results

Question (Ariel/Yoni): *which parts of the ImageNet-pretrained DINOv2 backbone should we fine-tune to adapt the model to fMRI? Everything? Only the last few transformer blocks? Nothing except the input adapter we introduce?*

Shared setup across the three runs: HCP-only dataset, T=1200, temporal_kernel=20, base_lr=3e-4, effective batch 16 (2 x 8 grad accumulation), 20 epochs, init = official DINOv2 ViT-S/14 reg4 ImageNet checkpoint filtered to drop pos_embed and patch_embed.

Only the **parameter-freezing policy** differs between runs.

1. The three strategies

Strategy	Trained components	Trainable params	Description
A – Full fine-tune (baseline)	full backbone + heads	~22 M	Official DINOv2 behaviour: no freezing
B – Partial freeze (freeze_last3)	patch_embed.* + blocks.9–11 + norm + heads	~3.6 M	The first 9 ImageNet blocks are frozen; the last 3 + final LayerNorm are fine-tuned
C – Freeze except input (freeze_fmri)	patch_embed.* + heads	~300 k	The entire ImageNet backbone is frozen. Only the 3D+1D adapter we added is trained

For each strategy we keep the **best checkpoint** among the available iters (11999, 17999 for the two freeze runs; 5999, 8999 for the baseline).

2. Results – HCP probes (the pretraining dataset)

Label	Metric	A (full FT)	B (freeze last3)	C (freeze fmri)	Winner	C vs A
Sex	AUC, higher is better	0.751	0.696	0.844	C	+0.093
BrainVol	MAE, lower is better	116,179	117,744	99,455	C	-14 %
GrayMatterVol	MAE	34,762	34,815	29,274	C	-16 %
Age	MAE	3.044	3.056	2.931	C	-3.7 %
FluidIntel	MAE	4.047	4.035	4.003	C	-1.1 %
ProcSpeed	MAE	16.343	16.401	16.207	C	-0.8 %
WorkingMem	MAE	10.442	10.410	10.411	B (tie with C)	~ -0.3 %

==> Strategy C (freeze except input) wins on 6 of 7 HCP labels.

3. Results – ADNI probes (transfer dataset)

The baseline (A) ADNI numbers come from the canonical probe on checkpoint 144327 (HCP-only T=1200, iter 8999) – **same methodology** as the freeze probes (5-fold CV, LogReg C=1.0, sklearn).

Label	Metric	A (full FT)	B (freeze last3)	C (freeze fmri)	Winner	C vs A
Sex	AUC	0.653	0.696	0.813	C	+0.160
Degradation3Y	AUC	0.528	0.468	0.544	C	+0.016
Degradation1Y	AUC	0.581	0.543	0.559	A	(-0.022)
Degradation2Y	AUC	0.533	0.456	0.533	A tied with C	~ 0
CDR	AUC	0.542	0.522	0.532	A	(-0.010)
MMSE	MAE	2.478	2.426	2.750	B	(+11 %)
Age	MAE	5.113	4.993	5.703	B	(+12 %)

A more nuanced picture on ADNI: - **Anatomical** (Sex): strategy C wins by a wide margin (+0.16 AUC). - **Demographic** (Age) and **cognitive** (MMSE): strategy B (partial FT) wins. - **Clinical progression** (Degradation1Y, CDR): the full FT baseline (A) is slightly better, but the gap is **within fold noise** (+/- 0.05 to +/- 0.10 std on 5-fold AUC).

4. Three key findings

4.1 On the pretraining dataset (HCP), freeze-except-input beats everything

6 of 7 HCP labels are won by strategy C. On **Sex** the margin is large: 0.844 vs 0.751 baseline = **+9.3 AUC points**. On **BrainVol** and **GrayMatterVol** the MAE drops by 14-16 %.

Interpretation: the generic visual features learned by DINOv2 on ImageNet (edges, textures, shapes) transfer **remarkably well** to fMRI once the input adapter is trained appropriately. **Full backbone fine-tuning degrades those features** by over-specialising on the small HCP set (~1k scans).

4.2 Strategy B (partial freeze) is the worst on HCP

Fine-tuning only the last 3 blocks creates an **internal discontinuity**: blocks 0-8 stay frozen at ImageNet weights while blocks 9-11 adapt to fMRI. The low-level (ImageNet) vs high-level (fMRI) feature mismatch hurts the overall quality more than either extreme.

4.3 On ADNI transfer: strong anatomical, clinical signal still in noise

For Sex on ADNI (anatomical, n=812, ~50/50 balanced), strategy C beats A by +0.16 AUC – **clear and robust**. For clinical labels (Degradation, CDR), all runs sit between 0.52 and 0.58 AUC with +/- 0.05 to +/- 0.10 std on 5-fold CV: differences are **within noise**. This is consistent with the earlier observation that the clinical signal is **data-limited** (~145 positive Degradation1Y cases).

5. Why strategy C is the right choice for the thesis

Argument	Detail
Robust HCP improvement	6/7 labels won, large gains on Sex / BrainVol / GrayMatterVol
Strong ADNI Sex improvement	+0.16 AUC vs baseline – well above the fold variance
Compute and parameter efficiency	300k trainable params vs 22M (x73 less) – faster training, lower overfitting risk
Clean interpretation	Matches the transfer-learning literature: “linear probing > full fine-tuning” on small datasets
Simple architectural story	“3D+1D input adapter + frozen ImageNet backbone” is a clear narrative for the defense

6. Caveats and limits

- The 2,999 / 5,999 / 8,999 intermediate checkpoints of the two freeze runs were not saved (PeriodicCheckpoint's `max_to_keep`), so the full training dynamics aren't visible.
 - `model_final` (iter 19,999) has not been probed yet for the two freeze runs – to be added for the final version of the table.
 - The ADNI baseline (A) comes from an earlier probe (before file renaming). The methodology is identical (canonical 5-fold LogReg) so the comparison is valid, but the same folds were not used.
 - Clinical labels remain **data-limited** (~145 positive Degradation1Y cases) – so differences between strategies on those labels are within noise. The bottleneck is the scarcity of positive cases, not the pretraining strategy.
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7. Reproducibility

- **Repo / branch:** <https://github.com/DanAbergel/dinov2> – branch `fmri-mixed-t140`
- **YAML configs:**
 - A: `dinov2/configs/train/fmri_vits.yaml` (with `dataset_path: HCP`, `fmri_temporal_size: 1200`, `fmri_temporal_kernel: 20`)
 - B: `dinov2/configs/train/fmri_vits_hcp_freeze_last3.yaml`
 - C: `dinov2/configs/train/fmri_vits_hcp_freeze_fmri.yaml`
- **One sbatch script for all three:** `slurm_jobs/run_dinov2_fmri.sh` – switch configs via `CONFIG_FILE=...`
- **Probes:** `slurm_jobs/probe_adni.sh` + `slurm_jobs/probe_hcp.sh`, auto-launched by `slurm_jobs/probe_3ckpts`
- **Summary table:** `python3 scripts/summarize_probes.py` – automatic comparison across all runs.
- **Freezing mechanism:** `apply_freeze_policy()` in `dinov2/train/train.py` (tagged # FMRI CHANGE), activated by `cfg.optim.freeze_pretrained` in `{"fmri_only", "fmri_plus_last_3", null}`.